

CampusEats – Service Design

2. Capabilities

1. Register User
2. Login User
3. Browse Menu
4. Search Food
5. Place Order
6. Make Payment
7. Track Order
8. Cancel Order
9. Update Menu
10. Prepare Food
11. Deliver Order
12. Send Notification
13. Manage Users

3. Service Design and Responsible Member

Service	Main responsibilities	Data owned	Member
User Service	Register, login, manage users	Users, User Accounts	Brajkshi Mandloi
Catalogue Service	Browse menu, search food, update menu	Vendors, Cafeterias, Menus, Food Items	Brajkshi Mandloi
Order Service	Place, cancel, track, prepare and deliver orders	Carts, Cart Items, Orders, Order Items, Deliveries	Brajkshi Mandloi
Payment & Notification Service	Make payments, send notifications	Payments, Notifications	Brajkshi Mandloi

4. Service Contracts

User Service

Operation	Input	Output	Errors
registerUser	user details	registration result	User already exists
loginUser	login details	login result	Invalid credentials
getUser	user reference	user information	User not found

Catalogue Service

Operation	Input	Output	Errors
browseMenu	menu criteria	menu items	Menu unavailable
searchFood	search criteria	matching food items	Invalid search
checkItem	item reference	item availability/details	Item unavailable

Order Service

Operation	Input	Output	Errors
createCart	user reference	cart	Invalid user
placeOrder	user and cart information	order confirmation	Empty cart; item unavailable; payment failure
trackOrder	order reference	order status	Order not found
cancelOrder	order reference	cancellation result	Order cannot be cancelled

Payment & Notification Service

Operation	Input	Output	Errors
createPayment	payment information	payment result	Payment failed
getPaymentStatus	payment reference	payment status	Payment not found
sendNotification	notification information	notification result	Notification failed

5. Full Specification – placeOrder

Purpose: Place a food order for a user from the selected cart items.

Inputs: User reference, cart information, delivery information, and payment information.

Outputs: Order confirmation, order status, and payment status.

Error cases: User not found, cart empty, item unavailable, invalid order, payment failure, or dependent service unavailable.

Details hidden from callers: The caller does not need to know the internal database tables, SQL queries, internal identifiers, transaction handling, or how the service stores and processes the order.

6. Service Validation

Service	Reachable	Self-contained	Has Contract	Independent	Loosely Coupled
User Service	Yes	Yes	Yes	Yes	Yes
Catalogue Service	Yes	Yes	Yes	Yes	Yes
Order Service	Yes	Yes	Yes	Yes	Yes
Payment & Notification Service	Yes	Yes	Yes	Yes	Yes

Validation note: Each service is intended to be reachable through its operations, owns its data, exposes a contract, avoids direct ownership of another service's data, and communicates through service operations rather than shared database tables.